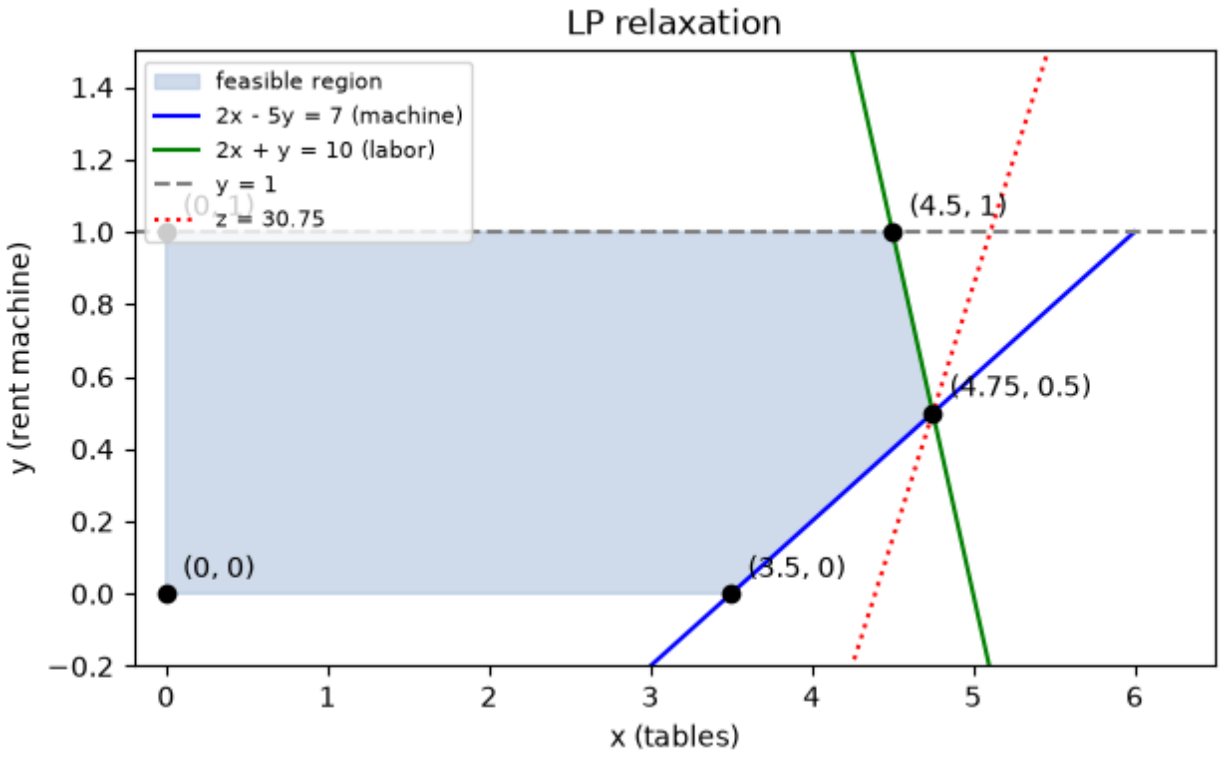
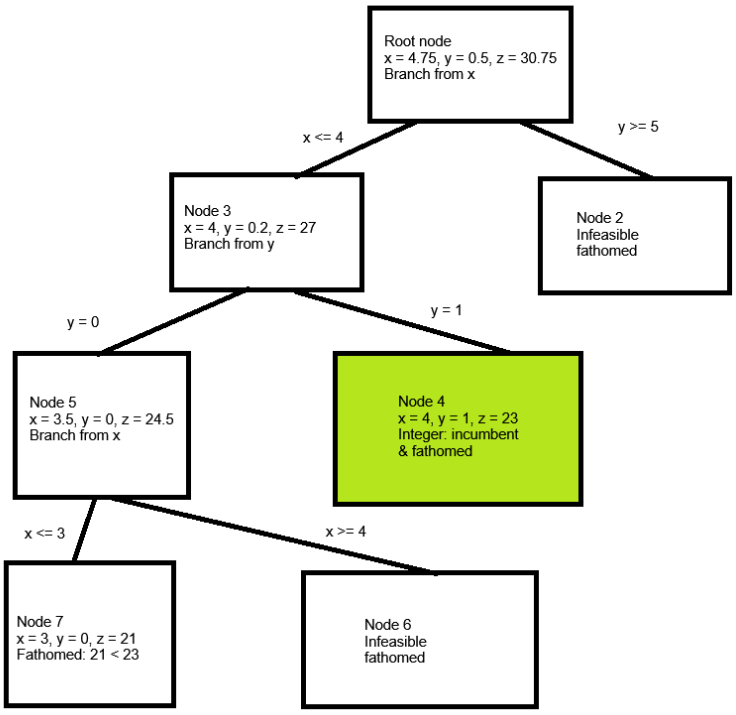
1.

a. The feasible region of the LP relaxation has corner points (0, 0), (3.5, 0), (4.75, 0.5), (4.5, 1), and (0, 1), with objective values 0, 24.5, 30.75, 26.5, and −5. The optimal LP solution is x = 4.75 and y = 0.5, with z = 30.75, where the machine time and labor lines cross.



b. Rounding to the nearest integers gives (5, 1), which needs 11 hours of labor, and rounding down gives (4, 0), which needs 8 hours of machine time, therefore neither point is feasible. Rounding an LP solution can leave the feasible region.

c. The root branches on x, the x ≥ 5 node is infeasible, and the x ≤ 4 node branches on y. The y = 1 node gives the integer solution x = 4, y = 1, z = 23, which becomes the incumbent. The y = 0 node (z = 24.5) branches on x again, where x ≥ 4 is infeasible and x ≤ 3 (z = 21) cannot beat 23. The optimal solution is to build 4 tables and rent the machine, for z = 23 ($2,300).



d. Branching on y fixes it at 0 or 1, which makes the full rent decision, so y is never branched on again below that node. Branching on x only adds a bound that cuts out the fractional values in between, so x can be branched on again further down, like with node 5.

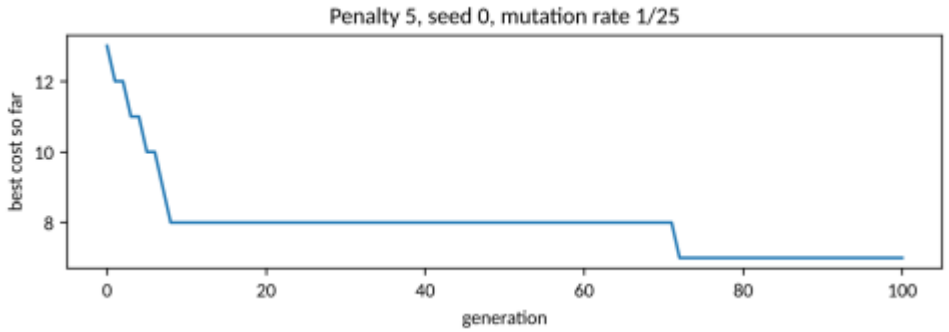
2.

a. Cost adds the number of towers to the penalty times ***count_uncovered*** and mutate flips each gene when ***rng.random()*** is below the rate. Every test value matches:

```
cost, no towers, penalty 5:    125   should be 125
cost, seven towers, penalty 5: 7   should be 7
cost, six towers, penalty 5:   16   should be 16
cost, six towers, penalty 1:   8   should be 8
mutate all towers, rate 0.5: [1, 1, 0, 0, 1, 0, 1, 0]   should start [1, 1, 0, 0, 1, 0, 1, 0]
```

b. 6 of the 10 runs (seeds 0, 1, 2, 5, 6, and 8) found an optimal solution of 7 towers with no uncovered regions, and the other 4 ended with 8 towers.

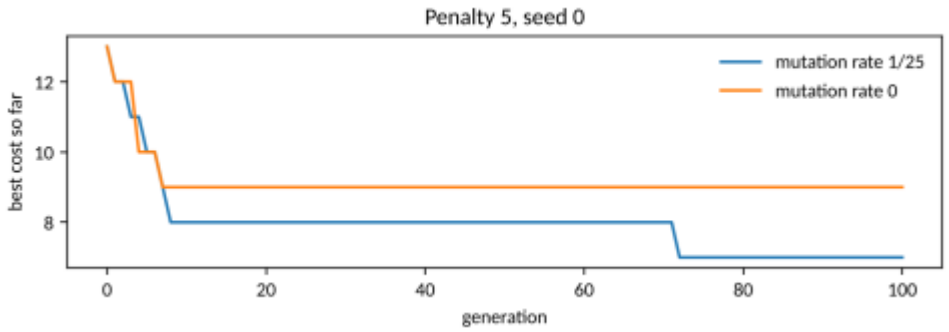
Seed ID	Towers	Uncovered	Regions
0	7	0	2, 5, 7, 14, 16, 20, 23
1	7	0	1, 4, 10, 12, 13, 21, 24
2	7	0	3, 6, 10, 13, 16, 20, 23
3	8	0	4, 5, 6, 7, 14, 17, 22, 25
4	8	0	1, 3, 5, 12, 14, 17, 20, 23
5	7	0	3, 6, 10, 13, 18, 21, 25
6	7	0	1, 4, 9, 12, 19, 21, 24
7	8	0	2, 4, 7, 12, 15, 17, 20, 23
8	7	0	3, 6, 10, 14, 17, 20, 24
9	8	0	3, 5, 6, 12, 14, 18, 21, 25



c. With no mutation, none of the runs found 7 towers and every run stopped improving by generation 12. Crossover only recombines genes already in the population, so once the population fills with copies of a few chromosomes it gets stuck. Mutation adds new genes and keeps the population diverse, which is what let seed 0 improve from 8 to 7 towers at generation 72.

Seed ID	Towers	Uncovered	Tower regions
0	9	0	3, 5, 6, 8, 15, 17, 18, 20, 25
1	9	0	1, 4, 8, 10, 11, 16, 19, 20, 23

2	8	0	1, 4, 7, 10, 13, 16, 21, 24
3	9	0	1, 3, 8, 10, 11, 13, 19, 22, 24
4	8	0	3, 5, 6, 8, 17, 19, 22, 25
5	9	0	2, 4, 7, 8, 15, 16, 19, 22, 24
6	8	0	1, 4, 8, 9, 11, 19, 22, 24
7	8	0	2, 4, 6, 10, 13, 17, 20, 23
8	9	0	2, 5, 10, 11, 13, 17, 20, 23, 25
9	8	0	2, 4, 6, 10, 12, 13, 21, 24



3.

a. The heuristic places the most constrained shifts first:
    1. List shifts from most to least conflicts with equals broken with lower shifts
    2. Take next shift
    3. Assigned to first hired guard with no shift conflict
    4. Hire a new guard only if no guard can take the shift
    5. Repeat 2-4 until shifts assigned

   Every shift is placed once with exactly one guard, and step 4 means it can always be placed. Step 3 only allows a guard with no conflicting shift, and a new guard has none, so the schedule is always feasible.

b. Ordering by conflicts gives 5, 2, 3, 4, 6, 7, 1, 8. Shift 5 hires Guard 1 and shift 2 joins it. Shift 3 conflicts with 2 and 5, so it hires Guard 2, and shift 4 joins Guard 2. Shift 6 conflicts with 5 and 4, so it hires Guard 3. Shift 7 joins Guard 2, shift 1 conflict with 2 and 3 so it joins Guard 3, and shift 8 joins Guard 1.

c. Shifts 1, 2, and 3 all conflict, so they need 3 different guards, and no schedule can use fewer than 3, which makes this schedule optimal. Since the heuristic is greedy, a bad order can hire extra guards. Choosing the next shift as the one whose conflicts already use the most different guards would improve it.